

MapFly: A Benchmark for Prior-Map-Guided Aerial Visual Navigation

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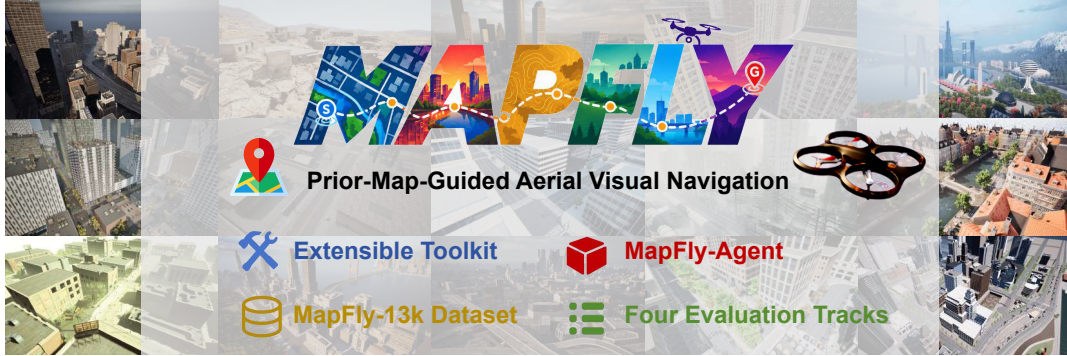


Fig. 1. **MapFly overview.** Prior-map-guided aerial visual navigation using annotated 2D prior maps and first-person observations. MapFly comprises an extensible toolkit, the MapFly-13K dataset, and MapFly-Agent, with four evaluation tracks defined by position cues and route guidance.

Abstract—Navigation intent for unmanned aerial vehicles (UAVs) is commonly specified through route instructions or target descriptions. However, users may know a destination’s map location without knowing the route or being able to describe its surroundings. Thus, we introduce MapFly, a benchmark for prior-map-guided aerial visual navigation, where UAVs navigate using first-person view (FPV) and annotated maps that provide global spatial context and specify navigation intent. MapFly defines four evaluation tracks by crossing static/live map-space position cues with route-free/route-assisted guidance, enabling controlled analysis of position cues and route guidance. It also provides an extensible toolkit for task creation, data collection, map rendering, and closed-loop evaluation. Using this toolkit, we construct MapFly-13K, a dataset of 13,300 validated episodes across 15 simulated outdoor scenes with matched map variants. We further develop MapFly-Agent as a reference policy for benchmark evaluation and input ablations. Experiments show substantial benefits from live position cues, but limited effective use of geographic map content. Meanwhile, navigation without live position cues remains challenging. We will publicly release the toolkit, dataset, and reference policy code upon publication.

I. INTRODUCTION

Autonomous navigation to a specified goal in unseen environments is a fundamental problem in embodied intelligence. As a prominent paradigm in embodied navigation, vision-and-language navigation (VLN) has been extended from ground agents to unmanned aerial vehicles (UAVs) [1], [2], [3], [4] operating in large-scale outdoor environments. Existing UAV VLN tasks typically specify navigation intent through detailed route instructions (Fig. 2, left) or target descriptions. In practice, however, a user may know where a destination lies on a geographic map without knowing the route or being able to describe its surroundings. This motivates a practical navigation setting in which two-dimensional (2D) prior maps, such as OpenStreetMap (OSM) maps or satellite imagery, provide global spatial context, while navigation intent is

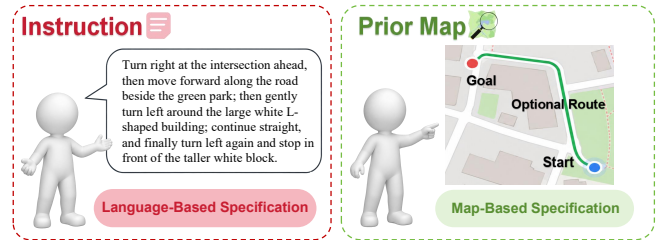


Fig. 2. **Navigation paradigms with different forms of intent specification.** Left: natural-language instructions providing a detailed route description. Right: a prior map with start and goal markers and an optional route overlay.

specified directly through start and goal markers and optional route overlays (Fig. 2, right).

Prior-map-guided navigation has been explored for ground robots and indoor agents [5], [6], [7], [8], showing that pre-existing maps can be effective for navigation. Extending this paradigm to outdoor UAV navigation, however, introduces additional challenges. UAVs need to combine map-guided global route planning with local motion planning based on first-person view (FPV). First, they must establish spatial correspondence between top-down geographic maps and first-person observations across large outdoor environments to connect global route plans with local flight decisions. Second, 2D geographic maps do not fully capture traversability at flight altitude, so UAVs must use local visual feedback to plan feasible movements.

In addition, prior maps in navigation applications for everyday human use often bundle several different sources of navigation information, including geographic content, live position cues, and route guidance. Evaluating navigation with a single annotated-map configuration therefore makes it difficult to disentangle their contributions to performance. This motivates two questions: (1) *How do map-space position cues and expert route guidance affect aerial navigation*

performance? (2) Within a given track, how effectively do navigation policies use geographic map content and FPV?

To address these questions, we introduce **MapFly**, a benchmark for prior-map-guided aerial visual navigation in simulated outdoor environments (Fig. 1). MapFly defines four evaluation tracks by crossing static/live map-space position cues with route-free/route-assisted guidance, enabling controlled comparisons of position cues and route guidance. Within a given track, controlled input ablations assess the contributions of geographic map content and FPV while preserving the other task inputs. To support task construction and evaluation, MapFly provides an extensible toolkit for expert trajectory generation, data collection, configurable map rendering, and closed-loop policy evaluation.

Using this toolkit, we construct **MapFly-13K**, comprising 13,300 validated navigation episodes across 15 outdoor scenes with matched prior-map variants. We further develop **MapFly-Agent** as a reference policy for benchmark evaluation and controlled input ablations. Experiments show that live map-space position cues substantially improve navigation performance. Expert route guidance improves navigation without live position cues, but navigation remains challenging even with this guidance. Input ablations reveal limitations in the evaluated policies’ effective use of geographic map content and highlight the importance of FPV for navigation.

Our contributions are threefold:

- 1) We introduce **MapFly**, a four-track benchmark for prior-map-guided aerial visual navigation, crossing static/live position cues with route-free/route-assisted guidance and providing an extensible toolkit for task creation, data collection, and closed-loop evaluation.
- 2) We construct **MapFly-13K**, a dataset of 13,300 validated navigation episodes across 15 outdoor scenes, with matched prior-map variants and Seen/Unseen evaluation splits.
- 3) We develop **MapFly-Agent** as a reference policy and provide systematic evaluations, baseline comparisons, and controlled input ablations to assess the effects of position cues and route guidance and the contributions of geographic map content and FPV.

II. RELATED WORK

A. Aerial Vision-and-Language Navigation

VLN has expanded from ground agents to aerial platforms. AerialVLN [1] introduced a UAV VLN task and a simulated urban navigation dataset. TravelUAV [2] incorporated realistic flight dynamics and continuous six-degree-of-freedom control with target-oriented trajectories, while OpenFly [3] scaled aerial VLN through automated multi-engine data generation. UAV-Flow [9] explores language-conditioned fine-grained flight control. CityNav [10] incorporates geographic semantic maps as auxiliary inputs for aerial navigation in real-world city representations, but navigation goals remain specified through language. In contrast, MapFly communicates navigation intent through direct annotations on prior maps rather than episode-specific language descriptions, using maps as both spatial priors and task interfaces.

B. Map Representations for Navigation

Metric maps. Classical navigation systems use metric maps that represent environment geometry and scale for localization and geometric planning. These maps are commonly constructed through simultaneous localization and mapping (SLAM), light detection and ranging (LiDAR) mapping, or three-dimensional reconstruction. However, constructing and maintaining accurate maps can be costly in large-scale environments [11].

Online map representations. Online maps organize observations gathered during navigation into structured spatial memory. VLFM [12] builds occupancy and language-grounded value maps for frontier exploration, while InstructNav [13] integrates semantic, trajectory, and action information into value maps. MapNav [14] constructs and updates an Annotated Semantic Map during navigation as spatial memory for a vision-language model (VLM)-based navigation policy.

Prior maps for navigation. Pre-existing maps have been explored in ground-robot navigation in several forms. CMN [5] investigates navigation using coarse and hand-drawn maps; FloNa [6] studies navigation to goals marked on indoor floor plans using egocentric RGB observations; SkeNa [7] uses abstract hand-drawn sketches; and FloorPlan-VLN [8] combines semantic floor plans with concise language instructions for navigation. These studies show how prior maps can support navigation despite being incomplete or visually distinct from egocentric observations. MapFly extends this setting to simulated outdoor UAV navigation using 2D geographic maps and FPV.

III. MAPFLY BENCHMARK

A. Task Formulation

In each episode, a UAV is tasked with navigating from a start position $\mathbf{p}_0$ to a goal region centered at $\mathbf{p}_g$, with a randomized initial heading. Success requires issuing a stop decision within the goal region without collision. At decision step t , its observation comprises an FPV RGB image I_t , a 2D prior map M_t , relative state s_t , and a fixed instruction q :

$$o_t = (I_t, M_t, s_t, q). \quad (1)$$

The state $s_t = [\Delta x_t^{(0)}, \Delta y_t^{(0)}, \Delta \psi_t^{(0)}]$ encodes displacement from the start in the episode-start body frame and yaw change from the initial heading. The map remains north-up regardless of the UAV’s heading. The fixed instruction defines the task and marker conventions, not an episode-specific route.

MapFly decouples high-level planning from low-level control. At each decision step, the policy outputs a stopping probability p_t^{stop} and a chunk of H waypoint increments:

$$A_t = (a_{t,1}, \dots, a_{t,H}), \quad (2)$$

$$a_{t,k} = (\Delta x_{t,k}, \Delta y_{t,k}, \Delta z_{t,k}, \Delta \psi_{t,k}).$$

Here, $a_{t,k}$ denotes the k -th increment in A_t , for $k = 1, \dots, H$. The components $\Delta x_{t,k}$, $\Delta y_{t,k}$, and $\Delta z_{t,k}$ are translations along the body-frame axes in meters, and $\Delta \psi_{t,k}$ is the yaw increment in radians. The increments use successive body frames, starting from the current UAV pose. Expert trajectories

(a) MapFly Toolkit



(b) MapFly-13K Data Generation

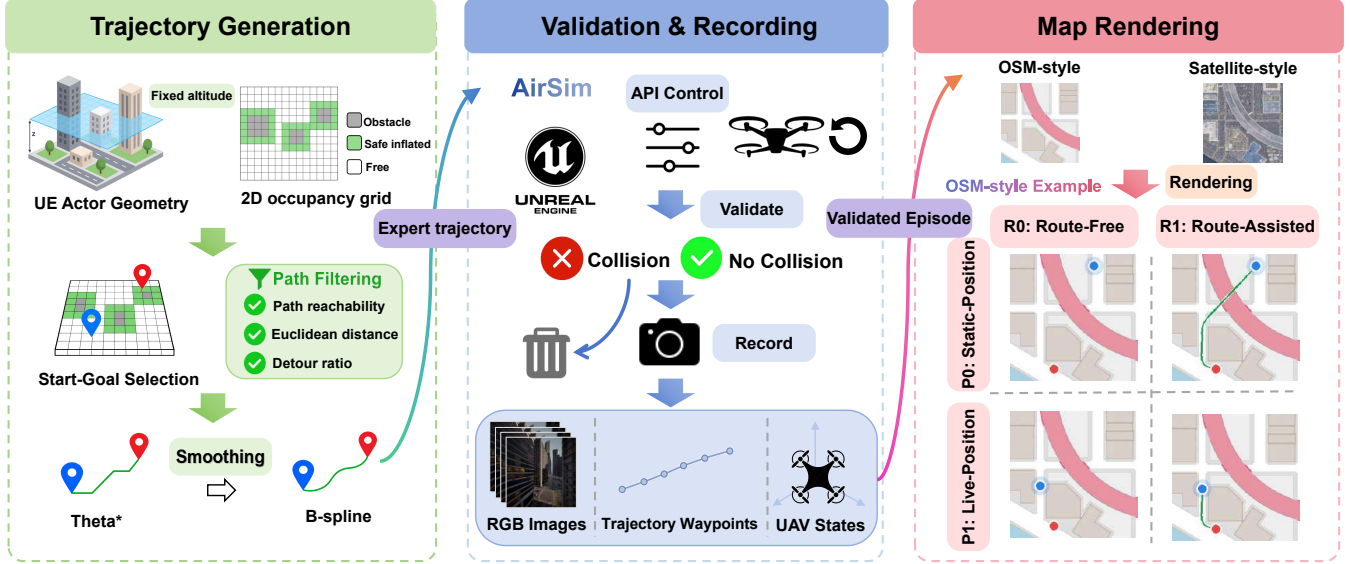


Fig. 3. **MapFly toolkit and dataset generation.** (a) Four toolkit modules supporting task construction, data collection, configurable map rendering, and policy integration and evaluation. (b) MapFly-13K generation: trajectory planning from scene geometry (left), flight validation and recording in AirSim (center), and matched map rendering (right). The OSM-style matrix crosses position cues (rows: P0/P1) with route guidance (columns: R0/R1). Planning geometry is used offline only.

maintain a fixed flight altitude within each episode, so the supervised vertical increments satisfy $\Delta z_{t,k} = 0$.

B. Simulation Platform and Benchmark Toolkit

MapFly is built on Unreal Engine (UE) and AirSim [15], using Cossys-AirSim [16] for UE5 scenes. UE handles scene modeling and rendering, while AirSim provides UAV flight simulation and control interfaces. The benchmark includes 15 outdoor scenes with diverse layouts and visual appearances, covering modern urban areas, industrial regions, and arid landscapes. Most scenes are urban, selected to challenge obstacle avoidance and route selection around built structures.

The toolkit comprises four modules (Fig. 3(a)): *task construction* for defining or sampling start-goal tasks with configurable episode parameters; *expert trajectory generation and data collection* for planning, flight validation, and synchronized recording; *configurable map rendering* for selecting map styles, geographic content, and task annotations; and *policy integration and closed-loop evaluation* for running custom policies through a common observation-action interface and recording metrics and rollouts. These modules support custom tasks, datasets, input configurations, and additional simulation scenes beyond MapFly-13K and the four benchmark tracks. For comparability, all experiments use fixed dataset splits, defined tracks or specified input ablations, and a common evaluation protocol.

As illustrated in Fig. 3(b), we use this toolkit to generate MapFly-13K through trajectory planning, flight validation and recording, and matched map rendering. We first construct a 2D

occupancy grid from UE Actor geometry at the predefined flight altitude and apply safety inflation. Next, start-goal pairs are sampled in free space and filtered by reachability and Euclidean distance. For each retained pair, Theta* [17] plans a collision-free path on the inflated occupancy grid. The resulting paths are further filtered by detour ratio and then smoothed using B-splines to produce continuous expert trajectories. These trajectories are subsequently executed in AirSim for flight validation. Episodes with collisions are discarded; the remaining episodes are retained with synchronized RGB images, trajectory waypoints, and UAV states. Finally, matched map variants are rendered from scene information and validated episode annotations, sharing the same start, goal, and expert trajectory without replanning.

C. Map Construction and Four-Track Design

Each map combines a geographic background with task annotations: a goal marker specifies the destination, a start or live-position marker provides a position cue, and an optional expert route provides guidance. We construct matched backgrounds in two styles: *OSM-style* maps represent scene geometry and semantic categories projected onto the horizontal plane, while *satellite-style* maps capture overhead appearance and texture. For the same episode and track, both styles share the same spatial coverage and task annotations. The viewport is north-up, determined by the start-goal pair, and fixed throughout each episode.

To examine how position cues and route guidance affect navigation difficulty, MapFly defines two factors, illustrated

by the OSM-style matrix in Fig. 3(b).

Position cues (P). The *Static-Position* setting (SP, P0) retains a static start marker, whereas the *Live-Position* setting (LP, P1) replaces it with a blue marker of the UAV’s live map-space position. This marker is externally supplied from simulator ground-truth position rather than estimated by the policy. Position markers do not encode heading; the geographic background and goal marker remain fixed. Both settings provide updated relative state s_t ; thus, P0 omits the live map-space marker rather than all position information.

Route guidance (R). The *Route-Free* setting (R0) omits the expert route, whereas the *Route-Assisted* setting (R1) overlays the complete expert route under P0 and the remaining expert route under P1. The remaining route runs from the point on the expert path nearest the current UAV position to the goal. Under P0, the route overlay remains static; under P1, it is updated using the same ground-truth position that supplies the live marker. The route-assisted tracks assess how expert guidance changes navigation difficulty and how well policies follow supplied routes, with route adherence evaluated separately from goal-reaching success.

Crossing these factors yields four benchmark tracks:

$$c = (P, R) \in \{0, 1\}^2, \quad (3)$$

where c denotes the track configuration, $P = 1$ indicates a live map-space position marker, and $R = 1$ indicates an expert route overlay.

- **Route-Free Static-Position Navigation (P0–R0)** uses static start and goal markers but no expert route.
- **Route-Free Live-Position Navigation (P1–R0)** provides live position grounding without route guidance.
- **Route-Assisted Static-Position Navigation (P0–R1)** displays the complete expert route with static start and goal markers, without a live position marker.
- **Route-Assisted Live-Position Navigation (P1–R1)** displays the remaining expert route together with live position grounding.

The four-track evaluation uses OSM-style maps and FPV to compare these position-cue and route-guidance conditions.

D. MapFly-13K Dataset and Splits

Generated with the MapFly toolkit, the **MapFly-13K** dataset contains 13,300 validated navigation episodes across 15 outdoor scenes. Each episode includes a start–goal task, an expert trajectory, synchronized first-person RGB observations, and matched prior-map variants. Navigation distance, measured as expert trajectory length, has a mean of 223.8 m; its distribution is shown in Fig. 4(a). The detour ratio is defined as trajectory length divided by the straight-line start–goal distance. Fig. 4(b) and (c) show the detour-ratio distribution and per-scene episode counts, respectively.

The dataset is divided into Train, Test Seen, and Test Unseen, containing 9,660 (72.6%), 2,140 (16.1%), and 1,500 (11.3%) episodes, respectively (Fig. 4(d)). Train and Test Seen draw from the same 12 scenes, with Test Seen containing held-out episodes. Test Unseen uses the remaining three scenes, which are excluded from training, to evaluate cross-scene

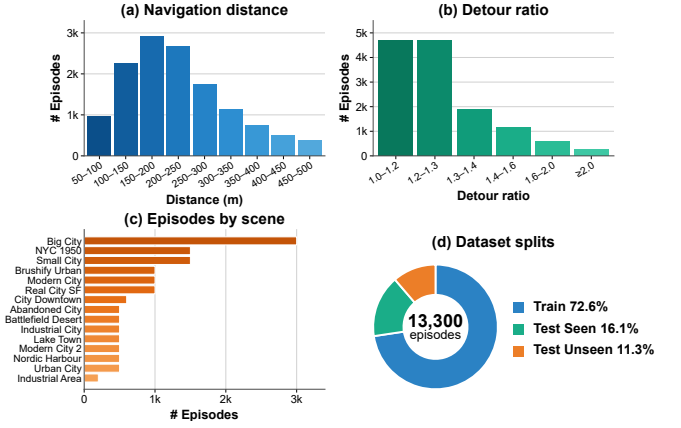


Fig. 4. **MapFly-13K statistics.** (a) Navigation distance (expert trajectory length); (b) detour ratio; (c) episodes by scene; (d) Train, Test Seen, and Test Unseen proportions.

generalization. All map variants of the same episode are assigned to the same split.

E. Closed-Loop Evaluation Protocol

The toolkit’s evaluation module connects policies through the common observation–action interface and executes the following closed-loop protocol in simulation. The UAV is initialized at the episode’s start position. At each decision step, the policy predicts a waypoint-increment chunk and a stopping probability using the interface in Sec. III-A. If $p_t^{\text{stop}} \geq 0.5$, the episode terminates at the current pose without executing the predicted chunk. Otherwise, the increments are converted to waypoints and executed in full by AirSim before the next observation is collected. A stop is successful if the UAV is within 10m of the goal. An episode terminates immediately upon collision, including during waypoint execution, or after 200 policy decisions.

We report success rate (SR), oracle success rate (OSR), navigation error (NE), success weighted by path length (SPL) [18], normalized dynamic time warping (nDTW) [19], and collision rate (CR). SR requires the policy to issue a stop decision within the goal region, whereas OSR measures entry into it at any point. Across all four tracks, adherence to the expert route is not required for SR and is evaluated separately by nDTW. NE is terminal Euclidean goal distance. SPL measures success and path efficiency using expert trajectory length as the reference. nDTW measures expert-trajectory agreement with a distance normalization threshold of 10m. CR is the fraction of episodes terminated by collision.

IV. METHOD

MapFly-Agent is a VLM-based reference policy adapted from StarVLA [20] for the MapFly benchmark.

A. Model Architecture

Using StarVLA’s QwenOFT implementation, MapFly-Agent combines Qwen3-VL-4B-Instruct [21] with a residual multilayer perceptron (MLP) action head adapted from OpenVLA-OFT [22] (Fig. 5). FPV I_t and map M_t are separate visual inputs. Relative state is normalized using training-set

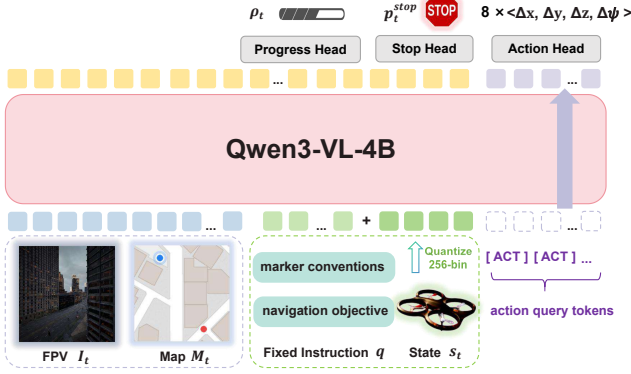


Fig. 5. **MapFly-Agent architecture using StarVLA’s QwenOFT implementation.** Qwen3-VL-4B processes FPV, a prior map, fixed instruction text, quantized state text, and eight action queries. The action head predicts eight waypoint increments. The stop and auxiliary progress branches share a feature mapping of the final action query’s hidden state.

statistics, quantized into 256 bins, and serialized as text. The backbone receives both visual inputs, a fixed task/map-convention prompt, state text, and $H = 8$ action queries:

$$\mathbf{h}_{t,1:N} = f_\theta(I_t, M_t, [\mathcal{P}(q); \mathcal{T}(s_t); \mathcal{Q}_H]), \quad (4)$$

where the causal VLM f_θ produces N final-layer hidden states. $\mathcal{P}$, $\mathcal{T}$, and $\mathcal{Q}_H$ denote the prompt, state serialization, and action queries, respectively; semicolons denote concatenation.

Two residual MLP blocks, shared across action queries, predict four-dimensional waypoint increments. Stop and auxiliary progress heads use a shared feature mapping of the final action query:

$$\begin{aligned} \hat{a}_{t,k} &= g_{\text{act}}(\mathbf{h}_{t,k}^{\text{act}}), \quad k = 1, \dots, H, \\ \hat{p}_t^{\text{stop}} &= g_{\text{stop}}(\mathbf{h}_t^{\text{end}}), \\ \hat{\rho}_t &= g_{\text{prog}}(\mathbf{h}_t^{\text{end}}). \end{aligned} \quad (5)$$

Here, $\mathbf{h}_{t,k}^{\text{act}}$ and $\mathbf{h}_t^{\text{end}} = \mathbf{h}_{t,N}$ are the k th and final action-query states; $\hat{\rho}_t$ is predicted progress. The stop/progress heads include the shared mapping and separate sigmoid outputs. All outputs are predicted in one pass without autoregressive action decoding; increments follow Sec. III-A. We call this implementation OFT, the default MapFly-Agent unless otherwise specified.

B. Training Objective

We jointly train the output heads and fully fine-tune the VLM on expert trajectories with

$$\mathcal{L} = \mathcal{L}_{\text{act}} + 0.05\mathcal{L}_{\text{stop}} + 0.5\mathcal{L}_{\text{prog}}. \quad (6)$$

$\mathcal{L}_{\text{act}}$ is masked ℓ_1 loss on normalized, non-padded increments. $\mathcal{L}_{\text{stop}}$ is binary cross-entropy on stop logits with positive-class weight 10; positive labels use an 8m goal radius, versus 10m for evaluation. $\mathcal{L}_{\text{prog}}$ is mean squared error for the completed fraction of reference-path length. On all tracks, these offline progress labels are used only as auxiliary supervision and are not provided as policy inputs. No language-modeling loss is used.

At inference, waypoint increments are unnormalized, execution and stopping follow Sec. III-E, and progress predictions are unused.

V. EXPERIMENTS

We first compare policies on P1–R0 with OSM-style maps and FPV, analogous to humans navigating with a digital map: the current position and destination are shown, but no route is prescribed. We then evaluate MapFly-Agent across all four tracks, ablate map representation, geographic background, and FPV, and analyze qualitative rollouts.

A. Experimental Setup

1) *Implementation Details:* MapFly-Agent is trained with a batch size of 32 on eight NVIDIA A100 GPUs and evaluated using the protocol in Sec. III-E. Both FPV and map inputs have a resolution of 224×224 pixels. All four-track comparisons retain FPV, and all reported configurations retain relative state. A separate policy is trained for each track or input configuration and evaluated with the corresponding inputs. Thus, input ablations compare separately trained policies, rather than test-time masking of a single policy.

2) *Baselines:* Our baselines include Seq2Seq (a sequence-to-sequence model) [23], CMA (a cross-modal attention model) [24], the Visual Navigation Transformer (ViNT) [25], UAV-Flow [9], and π_0 [26], adapted to the MapFly observation and waypoint interface. Seq2Seq and CMA use adaptations of the AerialVLN implementations [1]. All baseline comparisons use OSM-style maps and FPV on P1–R0. We also evaluate *MapFly-Agent (GR00T)*, which uses the same Qwen3-VL-4B backbone architecture but adopts StarVLA’s GR00T-style action architecture [27]. Unlike OFT’s text-based state input, this variant uses a dedicated numerical state encoder to condition the action model alongside VLM features. It retains the same stop and progress heads and their supervision.

3) *Evaluation Metrics:* We report SR, OSR, NE, SPL, nDTW, and CR on Test Seen and Test Unseen, as defined in Sec. III-E. In Tables I–III, NE is in meters; SR, OSR, and CR are percentages; and SPL and nDTW use a 0–100 scale. Arrows indicate better performance.

B. Main Results

1) *Route-Free Live-Position Policy Comparison:* Table I compares policies on P1–R0 with OSM-style maps and FPV. VLM-based policies—UAV-Flow, π_0 , and both MapFly-Agent variants—achieve higher SR in both splits than the non-VLM-based policies Seq2Seq, CMA, and ViNT. MapFly-Agent (OFT) leads across all six metrics in both splits, achieving 86.6% Seen SR and 62.7% Unseen SR, exceeding the best external baseline in each split by 32.8 and 34.5 percentage points, respectively.

Compared with MapFly-Agent (GR00T), OFT improves SR by 27.0 percentage points in both splits. The variants use the same VLM backbone architecture but differ in action architecture and state conditioning: OFT includes state in the VLM prompt, whereas GR00T conditions the action model

TABLE I

POLICY COMPARISON ON ROUTE-FREE LIVE-POSITION (P1-R0) WITH OSM-STYLE MAPS AND FPV. BOLD MARKS COLUMN-BEST VALUES.

Policy	Test Seen						Test Unseen					
	SR $\uparrow$	OSR $\uparrow$	NE $\downarrow$	SPL $\uparrow$	nDTW $\uparrow$	CR $\downarrow$	SR $\uparrow$	OSR $\uparrow$	NE $\downarrow$	SPL $\uparrow$	nDTW $\uparrow$	CR $\downarrow$
Seq2Seq [23]	0.9	1.2	166.7	0.8	3.9	78.1	1.8	1.9	162.5	1.7	5.4	65.9
CMA [24]	1.4	1.5	166.8	1.1	4.5	86.2	1.3	1.3	168.6	1.1	5.9	78.3
ViNT [25]	45.6	51.6	50.8	40.6	46.6	34.0	17.2	23.1	72.7	15.7	24.7	56.4
UAV-Flow [9]	53.8	56.4	44.2	49.7	57.0	13.2	28.2	30.8	64.3	26.6	35.9	32.0
π_0 [26]	52.9	58.4	57.8	47.0	51.1	17.0	28.2	31.1	68.7	26.3	34.7	35.9
MapFly-Agent (GR00T)	59.6	65.0	33.8	52.8	54.6	17.0	35.7	42.9	45.4	32.9	41.9	38.2
MapFly-Agent (OFT)	86.6	87.9	17.4	78.4	78.7	6.1	62.7	65.3	36.9	58.2	55.6	26.5

TABLE II

MAPFLY-AGENT (OFT) ACROSS THE FOUR BENCHMARK TRACKS, USING OSM-STYLE MAPS AND FPV. BOLD MARKS COLUMN-BEST VALUES.

Track			Test Seen						Test Unseen					
			SR $\uparrow$	OSR $\uparrow$	NE $\downarrow$	SPL $\uparrow$	nDTW $\uparrow$	CR $\downarrow$	SR $\uparrow$	OSR $\uparrow$	NE $\downarrow$	SPL $\uparrow$	nDTW $\uparrow$	CR $\downarrow$
P0-R0	Static-Position	Route-Free	34.6	40.5	63.8	31.4	55.8	7.5	7.1	11.3	78.4	6.7	32.2	14.8
P1-R0	Live-Position	Route-Free	86.6	87.9	17.4	78.4	78.7	6.1	62.7	65.3	36.9	58.2	55.6	26.5
P0-R1	Static-Position	Route-Assisted	59.1	70.8	25.6	53.1	77.8	4.8	37.0	41.1	35.3	34.1	60.3	16.8
P1-R1	Live-Position	Route-Assisted	89.0	90.7	14.6	80.1	87.3	5.5	49.9	52.0	63.4	46.0	54.6	29.5

through a dedicated numerical state encoder. These results favor the OFT configuration as a whole, rather than isolating the effect of the action architecture.

Higher success rates do not always imply lower collision rates: on Unseen, GR00T achieves higher SR but also higher CR than UAV-Flow and π_0 . Even OFT shows a 23.9-percentage-point Seen-Unseen SR gap, with CR rising from 6.1% to 26.5%, highlighting persistent challenges in cross-scene goal-reaching and collision avoidance despite live position grounding.

2) *Evaluation Across the Four Benchmark Tracks*: Table II reports the evaluation of separately trained MapFly-Agent policies across the four tracks, retaining OSM-style maps, FPV, and relative state. In the route-free comparison, live position grounding increases Unseen SR from 7.1% (P0-R0) to 62.7% (P1-R0), a gain of 55.6 percentage points. Because both tracks receive start-relative state, this comparison measures the benefit of explicitly displaying map-space position for the reference policy, rather than isolating visual localization ability. Without the live marker, the policy must relate its available observations and start-relative state to the prior map.

With live position cues, route assistance improves Seen SR from 86.6% to 89.0% but reduces Unseen SR from 62.7% to 49.9%. Likewise, nDTW increases on Seen but decreases slightly on Unseen, indicating that the gains in goal-reaching and route adherence do not generalize across scenes. We hypothesize that multiple feasible routes may exist, and guidance along a single expert route may interfere with the policy’s navigation decisions in unseen scenes.

Without live position cues, route assistance improves goal-reaching and expert-trajectory agreement in both splits: Seen/Unseen SR rises from 34.6%/7.1% to 59.1%/37.0%, while nDTW increases from 55.8/32.2 to 77.8/60.3. However,

SR remains below that of Route-Free Live-Position navigation in both splits, indicating that route guidance does not fully close the success-rate gap for this policy.

C. Ablation Studies

Following Sec. V-A, Table III examines input contributions primarily on P1-R0, using OSM-style maps with FPV as the reference. For OFT, we vary map representation, geographic background, or FPV individually, retaining live position and goal markers and relative state. We repeat the background ablation with GR00T on P1-R0 to check cross-architecture consistency and evaluate both policy variants on P0-R0 to assess whether its effect extends to navigation without live map-space position cues.

1) *Map Representation*: Replacing OSM-style maps with satellite-style maps changes Seen/Unseen SR only modestly, from 86.6%/62.7% to 86.8%/64.9%, while reducing Unseen NE and CR. Both styles retain geographic content and identical position-goal markers, so this comparison does not isolate the contribution of the geographic background.

2) *Geographic Background under Live Grounding*: To isolate the contribution of geographic background, we remove buildings, roads, and other geographic content from the OSM reference while preserving map extent, orientation, scale, and live position-goal geometry. FPV and relative state are unchanged. Removing the background increases Seen SR from 86.6% to 88.3% and Unseen SR from 62.7% to 71.7%. On Unseen, NE decreases from 36.9 m to 29.1 m, SPL rises from 58.2 to 65.9, and CR falls from 26.5% to 17.5%.

These results show that, for this policy under live position grounding, the markers-only input yields higher navigation success rates than the OSM input. This suggests that, when using the map, the policy may rely only on the relative positions of the live position and goal markers, rather than

TABLE III

MAPFLY-AGENT INPUT ABLATIONS ON ROUTE-FREE TRACKS. P1-R0: ROUTE-FREE LIVE-POSITION; P0-R0: ROUTE-FREE STATIC-POSITION. W/O: WITHOUT. BOLD MARKS COLUMN-BEST VALUES WITHIN EACH TRACK-POLICY GROUP.

Track	Policy	Input condition	Test Seen						Test Unseen					
			SR $\uparrow$	OSR $\uparrow$	NE $\downarrow$	SPL $\uparrow$	nDTW $\uparrow$	CR $\downarrow$	SR $\uparrow$	OSR $\uparrow$	NE $\downarrow$	SPL $\uparrow$	nDTW $\uparrow$	CR $\downarrow$
P1-R0	OFT	OSM + FPV	86.6	87.9	17.4	78.4	78.7	6.1	62.7	65.3	36.9	58.2	55.6	26.5
		Satellite + FPV	86.8	87.9	17.6	78.5	77.8	6.8	64.9	68.5	31.2	59.9	57.3	21.7
		Markers-only + FPV	88.3	88.6	16.5	79.8	78.1	5.8	71.7	72.9	29.1	65.9	59.6	17.5
		OSM w/o FPV	55.7	57.8	59.7	50.0	51.2	37.3	33.3	39.1	61.5	30.7	35.4	53.7
	GR00T	OSM + FPV	59.6	65.0	33.8	52.8	54.6	17.0	35.7	42.9	45.4	32.9	41.9	38.2
		Markers-only + FPV	73.7	80.1	25.3	66.0	66.3	10.5	51.3	59.3	35.5	46.8	47.7	23.7
P0-R0	OFT	OSM + FPV	34.6	40.5	63.8	31.4	55.8	7.5	7.1	11.3	78.4	6.7	32.2	14.8
		Markers-only + FPV	24.3	34.7	73.3	21.9	49.4	8.2	9.2	11.9	83.2	8.7	30.9	16.5
	GR00T	OSM + FPV	19.8	36.9	102.6	17.5	37.7	8.4	1.6	7.6	152.5	1.5	9.6	9.5
		Markers-only + FPV	8.1	26.8	125.4	7.3	26.7	7.3	1.1	9.5	158.6	1.0	10.8	8.1

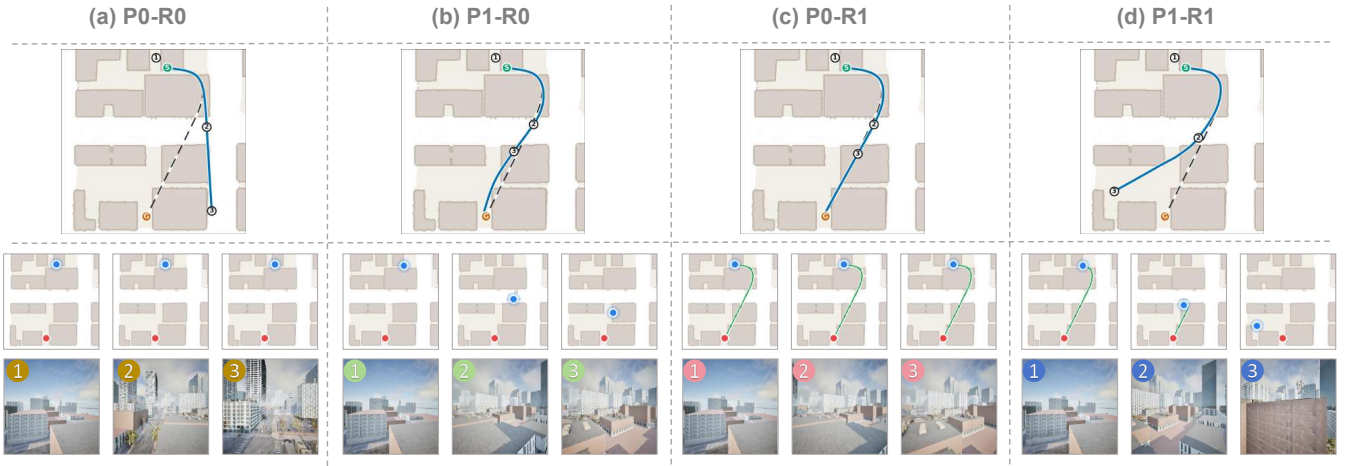


Fig. 6. MapFly-Agent (OFT) rollouts across the four benchmark tracks. Columns (a)–(d): P0-R0, P1-R0, P0-R1, and P1-R1, respectively. Top: expert (dashed) and executed (blue) trajectories, with start S and goal G. Middle/bottom: map/FPV inputs at the numbered trajectory samples 1–3. Dashed expert trajectories in the top row are for comparison; route overlays are supplied to the policy only on R1.

effectively exploiting geographic map structure. The findings point to a limitation in the policy’s ability to translate geographic map information into higher navigation success rates, rather than a lack of value in map structure itself.

GR00T also benefits from background removal: SR increases from 59.6% to 73.7% on Seen and from 35.7% to 51.3% on Unseen, with all remaining metrics improving in both splits (Table III). The consistent trend across both architectures suggests that the benefit is not specific to OFT.

3) *First-Person View*: The markers-only results retain FPV and therefore do not establish the sufficiency of map-space cues alone. To assess the contribution of FPV, we separately remove it from the OSM + FPV reference while retaining the geographic background, live position and goal markers, and relative state. Seen/Unseen SR drops from 86.6%/62.7% to 55.7%/33.3%, while CR rises from 6.1%/26.5% to 37.3%/53.7%. Navigation error also increases substantially in both splits. These results highlight the importance of FPV for goal-reaching and collision avoidance, suggesting that local visual feedback helps the policy perceive nearby obstacles and adjust its movements during flight.

4) *Geographic Background without Live Grounding*: The P0-R0 groups in Table III extend the comparison to static-position navigation for both OFT and GR00T, replacing the OSM background with a uniform-color background while retaining static start and goal markers, FPV, and relative state. Neither input includes a live-position marker or expert route. For OFT, unlike the live-position result, removing the background reduces Seen SR from 34.6% to 24.3%, with all other metrics also worsening. On Unseen, SR increases slightly from 7.1% to 9.2%, but NE rises from 78.4m to 83.2m, CR rises from 14.8% to 16.5%, and nDTW falls.

For GR00T, background removal also reduces Seen SR, from 19.8% to 8.1%, while NE rises from 102.6m to 125.4m; all remaining metrics worsen except CR. On Unseen, SR decreases from 1.6% to 1.1%, NE rises from 152.5m to 158.6m, and SPL decreases, although OSR, nDTW, and CR improve. The very low Unseen success rates limit conclusions about effective geographic map use across scenes.

These results suggest that geographic map structure may support navigation without live position cues, particularly in seen scenes. However, the mixed effects and low success

rates on unseen scenes suggest that the evaluated policies may still struggle to exploit this structure effectively across environments.

D. Qualitative Analysis

Fig. 6 compares four-track rollouts with a common start-goal layout. Without route guidance, P0–R0 (a) diverges, whereas P1–R0 (b) approaches the goal. With route guidance, P0–R1 (c) follows the expert trajectory more closely and approaches the goal, while P1–R1 (d) deviates from it. These rollouts illustrate the setting-dependent effects of route guidance discussed in Sec. V-B.2.

E. Limitations

The evaluated policies achieve low success rates under static-position cues, particularly in unseen scenes, highlighting limitations in map–vision correspondence and spatial reasoning without live map-space position cues. Further research is needed to develop stronger navigation policies for these more challenging tracks without live position cues. In addition, the dataset uses fixed-altitude expert trajectories, limiting the study of autonomous altitude selection. Future extensions could incorporate variable-altitude trajectories.

VI. CONCLUSION

MapFly provides a four-track benchmark for prior-map-guided aerial visual navigation, supported by an extensible toolkit, the MapFly-13K dataset, and the MapFly-Agent reference policy. On the Route-Free Live-Position track, background removal increases success rates across both policy variants, whereas removing FPV lowers success rates and increases collision rates. Without live position cues, navigation remains challenging, and expert routes only partially close the success-rate gap with live-position navigation. These policy-specific findings highlight limitations in effective map utilization and the continued importance of visual feedback, motivating stronger map–vision correspondence and spatial reasoning.

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